ALGEBRA 2 · QUADRATIC FUNCTIONS & COMPLEX NUMBERS

Discriminant & Roots Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Discriminant & Roots Explorer** on Mr. K's Classroom: Algebra 2 → Quadratic Functions & Complex Numbers → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If a parabola never touches the x-axis, does the equation have real solutions?

   No. Its solutions are complex.

2 Explore use the tool and record what you see

Discriminant and roots

Equation	$b^2 - 4ac$	Number and type of roots	Roots
$x^2 - 2x - 3 = 0$	**16**	**2 real**	**3 and –1**
$x^2 - 6x + 9 = 0$	**0**	**1 real (repeated)**	**3**
$x^2 + 4 = 0$	**–16**	**2 complex**	**±2i**
$x^2 - 2x + 2 = 0$	**–4**	**2 complex**	**1 ± i**
$x^2 + 2x + 5 = 0$	**–16**	**2 complex**	**–1 ± 2i**
$2x^2 + 3x - 2 = 0$	**25**	**2 real**	$\frac{1}{2}$ **and –2**

3 Write the rule fill in each blank from what you observed

1. $b^2 - 4ac > 0$: **two real roots**. $b^2 - 4ac = 0$: **one repeated real root**. $b^2 - 4ac < 0$: **two complex roots**.

2. Complex roots of a quadratic with real coefficients come in **conjugate** pairs, a + bi and a – bi.

3. On the complex plane, conjugate roots are reflections over the **real** axis.

4. If the parabola opens up and its vertex is above the x-axis, the roots are **complex**.

4 Check no tool: show your work

1. Find the discriminant of $3x^2 - 4x + 5$ and describe the roots.

   –44; two complex roots

2. Solve $x^2 + 9 = 0$.

   $x = \pm 3i$

3. Solve $x^2 - 4x + 13 = 0$.

   $x = 2 \pm 3i$

4. For what c does $x^2 + 6x + c = 0$ have exactly one real root?

c = 9

5 Reflect

What does the discriminant tell you about the graph?

Its sign tells whether the parabola crosses the x-axis twice, touches it once, or misses it.